



AIoT revolution

The future of smart homes with next-gen motion recognition

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Artificial Intelligence of Things (AIoT) has been gaining widespread popularity, offering a seamless fusion of Artificial Intelligence (AI) and the Internet of Things (IoT). This convergence empowers devices to not only collect and transmit data but also to analyze and act on it in real time. Among its many applications, one of the most promising and transformative is its role in smart homes. AIoT-powered smart devices have redefined home automation by making them more intelligent,

responsive, and personalized. Now, with a groundbreaking new framework, WiFi-based motion recognition is set to take this innovation to the next level.

A team of researchers led by Professor Gwanggil Jeon from the College of Information Technology at Incheon National University, South Korea, has introduced a novel AIoT framework called the multiple spectrogram fusion network (MSF-Net) for WiFi-based human activity recognition. Their research, which was published in the

IEEE Internet of Things Journal, offers an advanced method of tracking human movement indoors using existing WiFi infrastructure. This development not only enhances the efficiency of smart homes but also has profound implications for security, healthcare, and energy optimization.

Why WiFi-based motion recognition matters?

In a typical smart home environment, devices rely on sensors such as cameras, infrared motion detectors, and wearables to monitor activity. However, these solutions often come with privacy concerns, high costs, and the need for additional hardware. WiFi-based motion recognition presents a compelling alternative. Since WiFi networks are already ubiquitous in modern homes, leveraging them for activity recognition eliminates the need for extra installations. Furthermore, this method ensures privacy since it does not rely on visual recordings but rather on analyzing how WiFi signals interact with human movement.

Accurate motion detection is a crucial element in making AIoT-powered smart homes truly intuitive. Recognizing